

SIX MCP Developer Guide — Swiss Hacks 2026

Getting Started

The SIX Financial Data MCP server exposes financial market data through a set of AI-friendly tools. You interact with it by connecting any MCP-compatible client (Claude Desktop, Claude Code, or any SDK that supports MCP) to the server URL and authenticating with your API key.

No mTLS certificate is required on your end.

MCP Configuration

Add the following entry to your MCP client configuration:

```
{
  "mcpServers": {
    "six-webapi-mcp": {
      "type": "streamable-http",
      "url": "https://ca-mcpwebapi-tools.nicepebble-599ed11f.westeurope.azurecontainerapps.io/mcp",
      "headers": {
        "Authorization": "Bearer <YOUR_API_KEY>"
      }
    }
  }
}
```

Authentication

You receive a unique API key from SIX. Pass it as a Bearer token in the **Authorization** header with every request.

```
Authorization: Bearer <YOUR_API_KEY>
```

The key is tied to a specific mTLS certificate on the server side — you do not need to manage certificates yourself.

Workflow

Follow this four-step pattern for every data request:

1. Find the instrument

```
find_instrument(text="Apple")
```

Returns valor numbers and MIC codes (e.g. valor **908440**, MIC **XNAS**).

2. Choose the right identifier format

Tool type	Identifier format	Example
Instrument tools	valor only	"908440"
Listing tools	valor + MIC (underscore)	"908440_XNAS"

3. Describe before executing

Always call the tool with `mode="describe"` first. The response shows available fields and a `root_type`. Then call `describe_type(name=<root_type>)` to see the full field list.

```
intraday_snapshot(mode="describe")
# → root_type: "IntradaySnapshot"
describe_type(name="IntradaySnapshot")
# → full field list with descriptions
```

4. Execute with validated fields

```
intraday_snapshot(
  mode="execute",
  listing_ids=["908440_XNAS"],
  fields=["dateTime", "last", "volume"]
)
```

Tools Available

Infrastructure Tools

Tool	Description
<code>get_datetime</code>	Returns the current date and time in a given timezone (default: Europe/Zurich). Useful for building date-range queries.
<code>describe_type</code>	Shows the full GraphQL SDL definition of a type — all fields with documentation. Call this after <code>mode="describe"</code> to explore available fields.

Search Tools

Tool	Description
<code>find_instrument</code>	Free-text search for instruments by name, ticker, or identifier. Returns valor numbers and MIC codes.
<code>criteria_search</code>	Structured search — filter instruments or listings by type, currency, country, and other criteria.

Instrument Tools

These tools accept `valors` (list of valor strings) and `fields`.

Tool	Description
<code>instrument_base</code>	Core reference data: identifiers, name, type, currency, issuer, markets.
<code>instrument_detail</code>	Extended attributes: bond coupons, option details, fund characteristics.
<code>instrument_markets</code>	All markets/exchanges where an instrument is listed. Use to find BC codes and venues.
<code>instrument_classification</code>	Classification codes: CFI, GICS, SSPA, risk measures, regulatory flags.
<code>instrument_symbology</code>	Cross-reference identifiers: ISIN, CUSIP, SEDOL, WKN, RIC, Bloomberg ticker.
<code>entity_base</code>	Issuer entity information: LEI, name, type, domicile.
<code>entity_classification</code>	Issuer industry/sector classification: GICS, ICB, NACE, SIC, Fitch.

Listing Tools

These tools accept `listing_ids` (list of "`valor_MIC`" strings) and `fields`.

Tool	Description
<code>listing_base</code>	Listing reference data: ticker, currency, trading status, price units.
<code>listing_symbology</code>	Listing-level identifiers: Bloomberg IDs, tickers, exchange codes.
<code>market_base</code>	Market information: MIC, name, location, trading hours, status.
<code>intraday_snapshot</code>	Current intraday market data: prices, volume, spreads, Greeks.
<code>end_of_day_snapshot</code>	Latest end-of-day settled data: close, settlement, daily metrics.
<code>end_of_day_history</code>	Historical end-of-day time series: OHLCV, yields, volatility, returns. Requires <code>date_from</code> .
<code>intraday_history</code>	Intraday historical data: OHLCV summaries by period or individual trade records.

Data Identifiers

Valor

The primary instrument identifier used by SIX. Pass as a string.

```
valors=["908440"]           # Apple
valors=["114621", "959184"] # Coca-Cola and Oracle (batch)
```

Listing ID (valor_MIC)

Used for listing-level data. Combine valor and MIC code with an underscore.

```
listing_ids=["908440_XNAS"] # Apple on NASDAQ
listing_ids=["114621_XNYS"] # Coca-Cola on NYSE
```

Tip: `find_instrument` returns the `mostLiquidMarket` MIC. If you get "No trading data available", call `instrument_markets` and look for venues flagged `BEST_VOLUME_OVERALL` or `HIGHEST_NUMBER_OF_TICKS`.
